Quiz 1

1- In Display, The red, green, and blue components of a pixel each use a pair of bits. These pairs are fed to three digital-to-analog converters (DAC's), which
a. choose the color
b. convert logical values into actual voltages
c. converts analog to digital signal

2- Which of the following is a primary advantage of using scalable vector graphics (SVG)?
a. It is limited to black and white graphics.
b. It is resolution-independent and scales without loss of quality.
c. It cannot be manipulated with CSS or JavaScript.
d. It only supports raster images.

3- Frame Buffer is :
a. A portion of flash-memory-register containing a bitmap that drives a video display.
b. A portion of Hard-disk containing a bitmap that drives a video display.
c. A portion of read-only-memory (ROM) containing a bitmap that drives a video display.
d. A portion of random-access-memory (RAM) containing a bitmap that drives a video display.

4- ——— Used for less details Images such as logos
a. Vector display
b. none
c. Raster display

5- Which of the following is a characteristic of vector graphics?
a. Always larger in file size than raster images
b. Composed of pixels
c. Made up of paths defined by mathematical equations
d. Limited to a fixed resolution

6- ————analyze pictures to derive descriptions
a. Image Processing
b. computer graphics
c. all
d. computer Vision

7- Using the direct coding method for Pixel color representation, How much memory is required for a 4-bit/pixel frame buffer of size 512x1024?
a. 4M byte
b. 256K byte
c. 2M byte
d. 512K byte

8- If a video is recorded at 60 FPS, but played back on a 30 Hz monitor, how many frames will be shown per second?
a. 30 FPS
b. 15 FPS
c. 120 FPS
d. 60 FPS

9- -------describes how objects appears (it defines attributes that describe color texture or shape of the object
a. all mentioned
b. mathematical model
c. physical model
d. Geometric model

10- Can we find a Raster Display in the real life?.
a. True.
b. False.

11- Graphics pipeline Defined as : It is a ..... that describes what steps a graphics system needs to perform to render a ..... to a .....
a. Design model - 3D scene - 2D screen
b. conceptual model - 3D scene - 2D screen
c. Mathematical model - 3D scene - 2D screen
d. conceptual model - 2D scene - 3D screen

12- Bitmap is :
a. Grid of pixels.
b. Grid of lines.
c. Grid of edges.
d. Grid of points.

13- POLYLINES is :
a. connected sequence of angles.
b. connected sequence of shapes.
c. connected sequence of straight lines.

14- A display that consists of a regular grid of pixels and is normally driven from a frame buffer called :
a. Raster Display.
b. LCD Display.
c. Vector Display.
d. LED Display.

15- Elements of Pictures Created in Computer Graphics
a. raster Image
b. polyline
c. all of the above
d. Text

16- What is a common characteristic of interactive graphics?
a. Static images displayed sequentially
b. Real-time rendering and user feedback
c. Use of traditional print media
d. Pre-rendered animations

17- Using the Look-Up Tables (LUTs) method for Pixel color representation, How much memory is required for a 4-bit/pixel frame buffer of size 512x1024 along with LUT of 6 byte wide
a.1M byte
b.4K byte
c.262144 byte
d.262240 byte.

18- Which of the following is an example of a non-interactive graphics system?
a. CAD software for 3D modeling
b. Video game
c. Virtual reality simulation
d. Animated movie

19- The instruction glutInitWindowPosition(100, 150); will apply x, and y coordinate from the origin pixel of display from:
a. right – up
b. right - down.
c. left - down.
d. left - up.

20- Concave Shape is:

a. Shape that have start point and end point not connected.

b. Polygon Shape Which have number of edges equal to number of angles.

c. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies outside the boundary of the shape.

d. Polygon Shape determined if a straight line can be drawn between two points in the shape that lies inside the boundary of the shape.

Quiz 2

1- clipping window is the viewport?
a. True
b. False

2- myReshape(GLsizei W, GLsizei H); function is used to----?
a. none
b. resize the view window
c. both display and view window
d. resize the display window

3- viewing port insure that the image will be displayed?
a. False
b. true

4- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the triangle of the vertex (10,10,10,10), (20,20,20),(30,30,30)
a. the triangle is clipped
b. the triangle is displayed
c. nothing appeared

5- for the following setting{glOrtho2D(0,2,1,1) and glViewport(40,60,360,240)} the image is displayed on ---?
a. (0,2,1,1) dimension
b. mapped to (0,2,1,1) dimension
c. mapped to (40,60,360,240) dimension
d. (40,60,360,240) dimension

6- the number of viewports(of dimension (10,20,50,40)) that can be displayed in a window size(40,100) . the viewports non overlapped and window positions (10,10)?
a. 40
b. 27
c. 5
d. non

7- ----------is where to be displayed?
a.Window
b.all
c.graphics
d.Viewport

8- The mapping of a two-dimensional, world-coordinate scene description to device coordinates is called ?
a.viewing transformation
b.display window
c.clipping window.
d.Mapped Normalized coordinates

9- If the user makes the screen window bigger the previous viewport still be viewed as previous without any change with resizing ?
a. False
b. True

10- to display the image in a short window the aspect ratio is-----?
a. Change the window size
b. Less than 1
c. Equal 1
d. Greater than 1

11- to display the image of dimension (30,30) in a display window five times on window diagonals ----?
a. For (int i=0,i<5,i++) glViewport(i,i,i*30,i*30)
b. non
c. For (int i=0,i<5,i++) glViewport(i,i,30+30*i,30+30*i)
d. For (int i=0,i<5,i++) glViewport(i,i,i+30,i+30)

12- to flip the window around the x axis you set the window of dimension 640X480 to-----?
a. setwindow(0,640,480,0)
b. setwindow(0,640,0,480)
c. setwindow((640,0,0,480)
d. setwindow(640,0,480,0)

13- to display the image in a tall window the aspect ratio is-----?
a. Equal 1
b. Change the window size
c. Less than 1
d. Greater than 1

14- to flip the window around the y axis you set the window of dimension 640X480 to-----?
a. setwindow((640,0,0,480)
b. setwindow(640,0,480,0)
c. setwindow(0,640,480,0)
d. setwindow(0,640,0,480)

15- the image you draw during the run an empty screen appears. This error is due to---?
a.all
b.viewing window size
c.glorthognal setting
d.display window size

16- we could set up a separate two-dimensional, viewing-coordinate reference frame for specifying the?
a.viewing transformation
b.clipping window.
c.Mapped Normalized coordinates
d.display window

17- ---------- is What to be displayed?
a. all
b. graphics
c. Viewport
d. Window

18- If the user makes the screen window bigger the previous viewport could still be viewed as previous with resizing ?
a.False
b.True

Quiz 3

1- From the disadvantage of DDA
a. 3_We cannot use the multiplication method.
b. 2 and 3
c. 1 and 2
d. 1_The floating-point arithmetic is time-consuming
e. 2_The accumulation of round-off error in successive additions of the floating-point increment.

2- however, the jaggies(stair step appearance) will be apparent to some extent even at the highest resolution possible with current technology, the jaggies(stair step appearance) will be apparent to some extent
a. false
b. true

3- In which algorithm is the decision parameter updated during the line drawing process?
a. Neither algorithm
b. Both algorithms
c. Bresenham's Midpoint Algorithm
d. Digital Differential Analyzer (DDA)

4- if the slop of line is small negative , we decide to change or not each time the value of -------
a. non
b. X value
c. Y value
d. decide each time step which to increment

5- The main advantage of Bresenham's algorithm over DDA is:
a.It is easier to implement
b.It can handle all types of lines equally well
c.It produces higher-quality lines
d.It requires less computational overhead

6- In Bresenham's algorithm, the decision parameter (P) is used to determine:
a. Whether to increment y or not
b. The slope of the line
c. The length of the line
d. The color of the line

7- In Bresenham's algorithm, when the decision parameter (P) is negative, the next pixel:
a. Is diagonally up and to the right
b. Is directly to the right
c. Is directly above
d. Is diagonally down and to the right

8- Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value is less than zero?
a. P_new = P + Δx - Δy
b. P_new = P + 2Δy
c. P_new = P + 2Δx
d. P_new = P + 2Δy - 2Δx

9- Which of the following is an advantage of the DDA over other line drawing algorithms?
a. It uses integer arithmetic exclusively
b. It requires less memory
c. It provides better quality lines
d. It is simpler to implement

10- Which of the following equations is used to update the decision parameter in Bresenham's algorithm if the decision parameter value was greater than zero?
a. P_new = P + 2Δy - 2Δx
b. P_new = P + 2Δy
c. P_new = P + Δx - Δy
d. P_new = P + 2Δx

11- The primary advantage of Bresenham's algorithm over other line drawing algorithms is:
a. It requires more memory
b. It uses floating-point arithmetic
c. It is faster and uses only integer calculations
d. It can draw curves easily

12- DDA Line Drawing Algorithm produce the following points for a line with endpoints (5,7) and (8,14).
a. (5,7) (5,8) (6,10)
b. (7,11) (7,12) (7,13)
c. (5,9) (6,10) (6,11)

13- What is the primary purpose of the Digital Differential Analyzer?
a. To manage memory allocation
b. To generate and display curves and lines
c. To perform arithmetic operations
d. To store large datasets

14- In the DDA algorithm, what is the significance of the variable m?
a. It determines the color of the line
b. It is used to calculate the length of the line
c. It represents the midpoint of a line segment
d. It indicates the slope of the line

15- Which of the following equations is used to calculate the first pixel position between endpoints (7,8) and (3,2)in DDA?
a. (x+1,y+m)
b. (x-1,y-m)
c. (x-(1/m),y-1)
d. (x+(1/m),y+1)
e. (x+1,y-1)

16- What is the initial value of the decision parameter (P) when drawing a line using Bresenham's algorithm with endpoints (20,10) and (30, 18)?
a. P = 6
b. P = 16
c. P = 4
d. P = 0

Quiz 4

1- By using Mid point algorithm to draw a circle with a radius of 9 units, along the circle octant in the 1st quadrant includes

a. (1,10), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)
b. (1,9), (2,10), (3,11), (4,11),(5,11),(8,12),(7,13)
c. (1,9), (2,9), (3,9), (4,9),(5,9),(8,8),(7,7)

2- If the points of an ellipse that lies in the first quad are (1,6), (2,6), (3,6), and(4,5) then the similar points on the second quad are:

a. (-1,6), (-2,6), (-3,6), (-4,5)
b. (-1,-6), (-2,-6), (-3,-6),(-4,-5)
c. (1,-6), (2,-6), (3,-6),(4,-5)

3- If the world coordinates of a point are (xW, yW) and the corresponding viewport coordinates are (xV, yV), what is the basic formula to convert from world coordinates to viewport coordinates?

a. XV= $XVmin+((XVmax-XVmin)/(XWmax-XWmin)* (XW-XWmin))$
b. XV=$(XVmax-XVmin)/(XWmax-XWmin)* (XW-XWmin)$
c. xv=xw/2
d. XV= $XVmin+(XVmax-XVmin)/(XWmax-XWmin)$

4- In the Midpoint Ellipse Algorithm, the two axes of the ellipse are defined by which of the following parameters?

a. Radius and Diameter
b. Height and Width
c. Major and Minor Axes
d. X and Y Coordinates

5- there is no limit to how big we can make the frame buffer and still maintain the refresh rate at 60 frames or more per second

a. false
b. true

6- What are the main parameters needed to implement the Midpoint Ellipse Algorithm?

a. The four corners of the ellipse
b. Radius and angle
c. Only the center point
d. Center coordinates and the lengths of the major and minor axes

7- What is the initial decision parameter used in the Midpoint Circle Drawing Algorithm for the first point of the circle?
a. p=r*r
b. p=r
c. p=1-r
d. p=r-2

8- What is the primary purpose of antialiasing in computer graphics?
a.To enhance texture details
b.To increase rendering speed
c.To reduce color depth
d.To minimize the visual distortion of edges

9- What is the purpose of the decision parameter in the Midpoint Ellipse Algorithm?
a. To decide which pixel to plot next
b. To determine the color of the ellipse
c. To set the rendering speed
d. To calculate the area of the ellipse

10- Which of the following techniques is commonly used for antialiasing?
a.Depth buffering
b.Texture mapping
c.Supersampling
d.Rasterization

11- ----------to model light and to perform simulations for animation
a.mathematical model
b.physical model
c.Geometric model
d.all mentioned

12- Bresenham Line Drawing for a line line with endpoints (10, 0) and (20, 8). This line has a slope of 0.8, includes
a.(12,2), (13,2), (14,3), (15,4)
b.(12,2), (13,3), (14,3), (15,4)
c.(12,2), (13,3), (14,4), (15,4)

13- Using the mid-point circle drawing algorithm, what's the initial point used to draw a circle with a radius of 8 units, along the circle octant in the 1st quadrant
a.(8,8)
b.(0,8)
c.(0,0)
d.(8,0)